

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Maximum - Minimum III: Multiple equality constraints

Dr Francois Ebobisse Bille

These notes are just for your own reading. They will not be discussed in class and hence are non-examinable.

In these notes, we study problems of the type

$$(P_3) \quad \begin{cases} \text{Finding Max/Min of } f(x, y, z) \\ \text{subject to } h_1(x, y, z) = 0 \\ \text{and } h_2(x, y, z) = 0. \end{cases}$$

where the functions $f, h_1, h_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ are continuously differentiable.

The approach here is through the method of Lagrange multipliers presented in the following theorem.

Theorem 0.1 (Third theorem of the method of the Lagrange multipliers)

Let $f, h_1, h_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ functions of class C^1 and let (a, b, c) be such that

$$h_1(a, b, c) = h_2(a, b, c) = 0 \text{ with } \nabla h_1(a, b, c) \times \nabla h_2(a, b, c) \neq (0, 0, 0).$$

If (a, b, c) is a solution to the problem (P_3) , then there exist $\lambda, \mu \in \mathbb{R}$ such that

$$\nabla f(a, b, c) = \lambda \nabla h_1(a, b, c) + \mu \nabla h_2(a, b, c) \Leftrightarrow \begin{cases} f_x(a, b, c) = \lambda h_{1x}(a, b, c) + \mu h_{2x}(a, b, c) \\ f_y(a, b, c) = \lambda h_{1y}(a, b, c) + \mu h_{2y}(a, b, c) \\ f_z(a, b, c) = \lambda h_{1z}(a, b, c) + \mu h_{2z}(a, b, c) \end{cases} \quad (1)$$

The constants λ and μ are called the *Lagrange multipliers*.

Proof: Since the functions h_1 and h_2 are continuously differentiable and

$$\nabla h_1(a, b, c) \times \nabla h_2(a, b, c) \neq (0, 0, 0),$$

it follows that the curve $\mathcal{C}$ intersection of the two level surfaces $h_1(x, y, z) = 0$ and $h_2(x, y, z) = 0$ is regular around the point (a, b, c) and can be described by a continuously differentiable function $\vec{r} : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^3$ such that

$$\vec{r}(0) = (a, b, c) \quad \text{and} \quad \vec{r}'(t) \neq (0, 0, 0).$$

Since the curve $\mathcal{C}$ lies on both surfaces it follows that

$$\vec{r}'(0) \perp \nabla h_1(a, b, c) \text{ and } \vec{r}'(0) \perp \nabla h_2(a, b, c) \Rightarrow \boxed{\vec{r}'(0) // \nabla h_1(a, b, c) \times \nabla h_2(a, b, c)}. \quad (2)$$

Now, if we assume that (a, b, c) is a point of minimum for the function f subject to both constraints $h_1(x, y, z) = 0$ and $h_2(x, y, z) = 0$, then we get in particular that the function $f \circ \vec{r}$ has minimum at the number $t = 0$. Therefore,

$$\frac{d}{dt}(f \circ \vec{r})(0) = 0 \Leftrightarrow \underbrace{\nabla f(\vec{r}(0)) \cdot \vec{r}'(0)}_{(\text{CR})} = 0 \Leftrightarrow \boxed{\nabla f(a, b, c) \perp \vec{r}'(0)}. \quad (3)$$

Therefore, we obtain from (2) and (3) that

$$\boxed{\nabla f(a, b, c) \perp \nabla h_1(a, b, c) \times \nabla h_2(a, b, c) \Leftrightarrow \begin{cases} \exists \lambda, \mu \in \mathbb{R} \text{ such that} \\ \nabla f(a, b, c) = \lambda \nabla h_1(a, b, c) + \mu \nabla h_2(a, b, c). \end{cases}}$$

and this achieves the proof of the theorem. ■

Remark 0.2 The condition

$$h_1(a, b, c) = h_2(a, b, c) = 0 \text{ with } \nabla h_1(a, b, c) \times \nabla h_2(a, b, c) \neq (0, 0, 0).$$

simply means that around the point (a, b, c) which lies on both level surfaces $h_1(x, y, z) = 0$ and $h_2(x, y, z) = 0$, the two surfaces cross each other. We recall that $\nabla h_1(a, b, c)$ and $\nabla h_2(a, b, c)$ are the normal vectors of the tangent planes to the level surfaces $h_1(x, y, z) = 0$ and $h_2(x, y, z) = 0$ at the point (a, b, c) respectively. Hence, the condition $\nabla h_1(a, b, c) \times \nabla h_2(a, b, c) \neq (0, 0, 0)$ means that both tangent planes are not parallel.

Example 0.3 Find the points of minimum for the function $f(x, y, z) := x^2 + y^2 + 2z^2$ subject the constraints $x + y = 1$ and $2x + y - z = 2$.

Solution: According to the method described in Theorem 0.1, we seek x, y, z, λ, μ such that

$$\begin{cases} 2x = \lambda + 2\mu \\ 2y = \lambda + \mu \\ 4z = -\mu \\ x + y = 1 \\ 2x + y - z = 2 \end{cases} \Rightarrow \begin{cases} x = \frac{1}{2}(\lambda + 2\mu) \\ y = \frac{1}{2}(\lambda + \mu) \\ z = -\frac{1}{4}\mu \end{cases}$$

which substituted in the last two equations gives the system of linear equations in λ and μ

$$\begin{cases} \lambda + \frac{3}{2}\mu = 1 \\ \frac{3}{2}\lambda + \frac{11}{4}\mu = 2 \end{cases} \Leftrightarrow \lambda = -\frac{1}{2} \quad \text{and} \quad \mu = 1.$$

Therefore, we obtain the point $\left(\frac{3}{4}, \frac{1}{4}, -\frac{1}{4}\right)$.

Another approach is to solve for instance y and z in terms of x from the constraints and substitute into the expression and find the point of minimum for that function of the single variable x defined by $g(x) = 4x^2 - 6x + 3$. However, notice that this approach does not always work for more complicated functions h_1 and h_2 .

Exercises 0.4

- Find the points of maximum and the points of minimum for each function subject to the given constraints.

$$(a) \quad f(x, y, z) := x + y; \quad \begin{cases} z = xy \\ x^2 + y^2 = 1 \end{cases}$$

$$(b) \quad f(x, y) := x - 2y + z; \quad \begin{cases} y^2 + z^2 = 1 \\ -x + y - z = 1 \end{cases}$$

- Find the highest and lowest points on the set

$$D := \{(x, y, z) \in \mathbb{R}^3: -x + y - z = -1 \text{ and } x = y^2 + z^2\}.$$

Hint: What are the functions f , h_1 and h_2 in this case?

The next set of notes will be on finding maximum and minimum on closed bounded domains!